SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

ANSWERS

Your five points are correct. To score well according to the rubric, you can present them as a **structured comparison with clear criteria, analysis, examples, and a final conclusion**.

1. Star vs Bus Topology

Criteria	Star Topology	Bus Topology	Analysis
Cost	Higher; requires switch and individual cables	Lower; uses a common backbone	Bus is better for very small, low-budget networks
Reliability	High; failure of one cable affects only one device	Low; backbone failure can affect the whole network	Star is more reliable
Performance	Better with switched Ethernet	Decreases as network traffic increases	Star provides better performance
Scalability	Easy to expand if switch has free ports	Limited	Star is more suitable for growing networks

Maintenance	Easy to identify faulty devices/cables	Fault location can be difficult	Star is easier to maintain

Example: A modern office generally uses star topology because individual computer failures do not bring down the entire network.

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2. Mesh vs Star Topology

Criteria	Mesh Topology	Star Topology
Fault tolerance	Very high	Moderate to high
Redundancy	Multiple paths available	Usually one path through central switch
Central device dependency	No single central device required	Depends on central switch
Cost	Very high	Lower
Complexity	High	Low
Suitable for	Critical networks	Offices, schools, labs

Analysis: Mesh topology is more fault tolerant because alternate paths can carry traffic when a link fails. Star topology is simpler and cheaper, but failure of the central switch can affect all connected devices.

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3. Bus vs Mesh Topology

Criteria	Bus Topology	Mesh Topology
Installation	Simple	Complex
Cabling	Low	Very high
Cost	Low	Very high

Maintenance	Difficult when backbone fails	Complex because of many links
Scalability	Limited	Limited by increasing complexity
Reliability	Low	Very high

For a full mesh containing **n devices**, the number of direct links is:

$$\text{Links} = \frac{n(n-1)}{2}$$

For example, with 5 devices:

$$\frac{5(5-1)}{2}=10\text{ links}$$

This shows why mesh becomes expensive and complex as the number of devices increases.

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4. Hybrid vs Star Topology

Criteria	Hybrid Topology	Star Topology
Scalability	Very high	High
Flexibility	Very high	Moderate
Cost	Depends on design	Moderate
Expansion	New network segments can be added	Limited by switch capacity
Management	More complex	Simple

Suitable for	Large organizations/campuses	Small and medium networks

Analysis: Hybrid topology is more flexible because different topologies can be combined according to requirements. For example, a university can use a **fiber mesh/ring backbone and star topology inside individual buildings**.

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5. Maintenance Comparison

Topology	Maintenance Difficulty	Reason
Star	Low	Individual faults are easy to isolate
Bus	Medium/High	Backbone problems can affect the entire network
Mesh	High	Many connections must be monitored
Hybrid	Medium/High	Different topology types require different maintenance approaches

Overall Analysis

- **Star:** Best balance of performance, reliability, cost, and maintenance for typical LANs.
- **Bus:** Simple and inexpensive but has limited scalability and reliability.
- **Mesh:** Provides the highest redundancy and fault tolerance but is expensive and complex.
- **Hybrid:** Provides the greatest flexibility and scalability for large networks, but administration is more complicated.

Conclusion

There is no single topology that is best for every situation. **Bus topology** is appropriate where low cost and simplicity are the main requirements. **Star topology** is generally preferred for modern LANs because it provides good performance, easy troubleshooting, and scalability.

Mesh topology is suitable for critical networks where high availability and redundancy are essential. **Hybrid topology** is ideal for large organizations such as universities because it combines the advantages of multiple topologies.

Key insight: The choice of topology should depend on **cost, reliability, performance, scalability, complexity, and maintenance requirements**, rather than choosing a topology based on only one factor.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data	Uses some examples	Limited examples	No supporting examples

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
		to support comparison			
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion